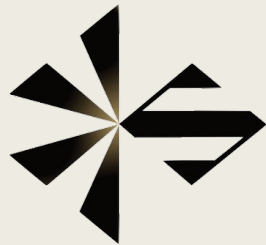


# Non-Traditional Machining Processes

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I) Conventional Machining vs Non-Traditional machining

II) Chemical and Electrochemical Processes

III) Mechanical Processes

IV) Thermal Processes

V) Synthesis and conclusion



## **IV) Thermal Processes**



# Thermal Processes



- *Electrical-Discharge Machining (EDM)*
  - Conventional vs Dry EDM
  - Plunge vs Wire EDM
- *Laser-Beam machining*
- *Electron-Beam machining*
- *Plasma Arc Machining*
- *(PAM vs Ion Beam Machining)*

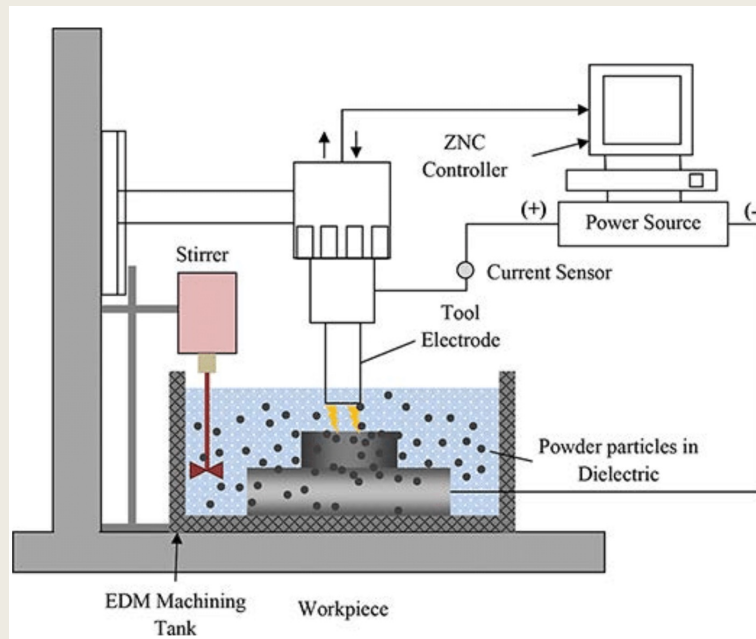




# Thermal Processes



## Electric Discharge Machining EDM





# Thermal Processes



## Conventional Electric Discharge Machining

vs

## Dry Electric Discharge Machining

Can be applied for conductive workpiece material only

Shape of the tool electrode must be in accordance with the intended feature

Mechanism of material removal is same

Spark generation occurs

Both die sinking EDM and wire EDM can be carried out either in conventional mode or in dry mode.

If process parameters are controlled properly then dry-EDM can offer similar MRR as conventional EDM.



# Thermal Processes



## Differences between conventional EDM and Dry EDM

| Conventional EDM                                                                                                                                       | Dry-EDM                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Suitable liquid (such as kerosene, deionized water, etc.) is used as dielectric fluid                                                                  | No liquid is used. Instead pressurized gas (like helium, nitrogen, argon or gas mixture) is used                |
| Spark is generated in a liquid medium                                                                                                                  | Spark is generated in a gaseous medium                                                                          |
| Dielectric liquid is usually reused for a longer duration                                                                                              | The dielectric gas is usually not recycled                                                                      |
| Liquid dielectric usually has high heat capacity                                                                                                       | Gaseous dielectric has low thermal capacity                                                                     |
| Cost of dielectric medium is comparatively high                                                                                                        | Dielectric medium is cheaper                                                                                    |
| Liquid dielectric can cause environment pollution, operator health hazard and fire                                                                     | No risk of environment pollution and fire, and at the same time it promotes operator's health                   |
| When ferrous material is machined by conventional EDM, the liquid dielectric prevents the atmospheric oxygen to come in contact with hot sparking zone | Chance of oxide layer formation on the hot machined surface is high, especially during machining ferrous metals |
| Liquid dielectric requires significantly higher voltage for breakdown                                                                                  | Breakdown voltage for gaseous dielectric is comparatively low                                                   |

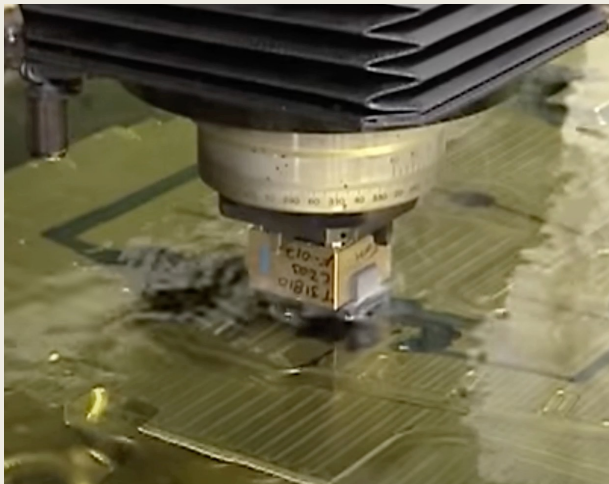




# Thermal Processes

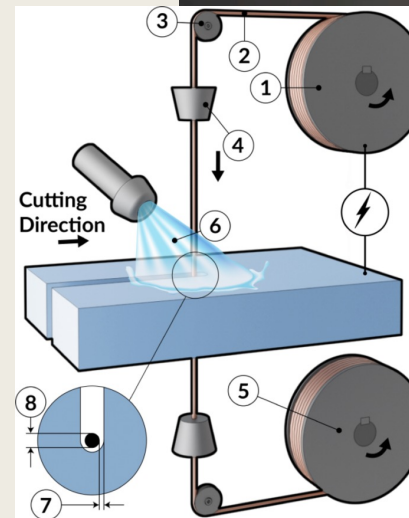
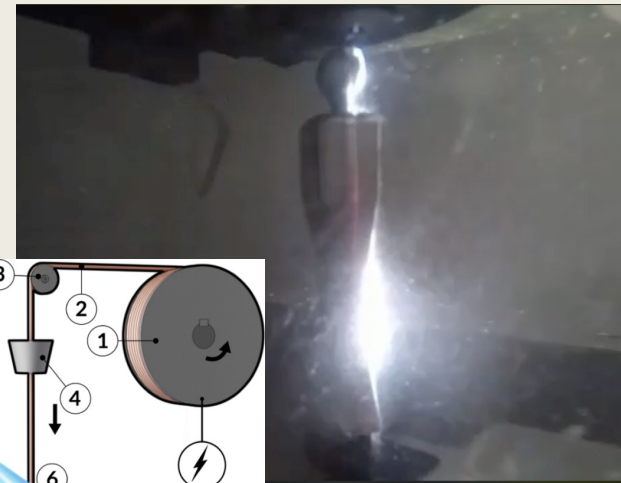


**Die Sinking (Plunge) EDM**



VS

**Wire EDM**





# Thermal Processes



## Die Sinking (Plunge) EDM

vs

## Wire EDM

Both are thermal energy based non-traditional machining processes

This mechanism is nothing but spark erosion or melting and vaporization of work metal

Suitable dielectric fluid is applied in both the cases to assist spark generation and debris removal

Presence of suitable potential difference between tool electrode (cathode) and workpiece (anode) is indispensably necessary for breakdown of dielectric fluid

None of them is associated with the production of burr, residual stress, etc.

None of them relies on mechanical properties of work material. However, electrical conductivity is one mandatory property of work material.



# Thermal Processes

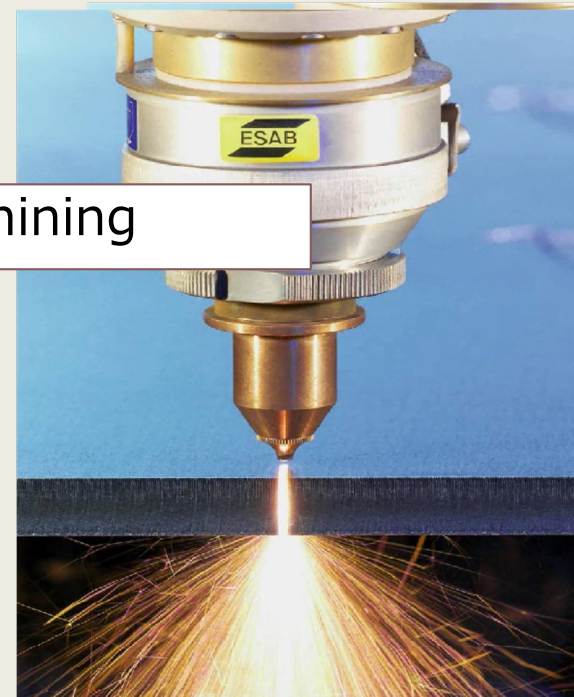
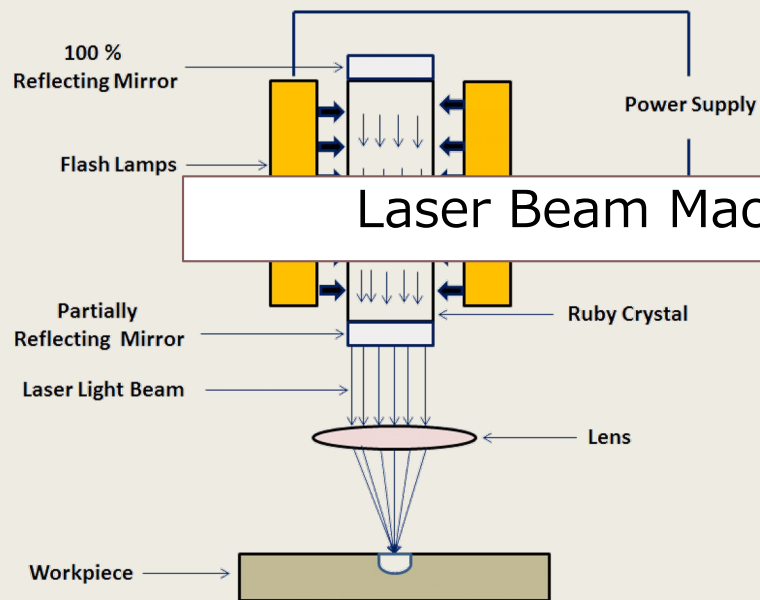


## Difference between Plunge (Die sinking) EDM and Wire EDM

| Die Sinking EDM                                                                                                     | Wire-EDM                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| A die made of conductive metal is used as tool electrode (cathode)                                                  | A long small diameter wire made of conductive metal is used as tool electrode or cathode                                        |
| The die can be given complex 3-D profile                                                                            | The wire always has a same simple geometry. So it is preferred for 2-D cutting only                                             |
| It can produce blind features                                                                                       | It is applicable for profile cutting by means of through holes only.                                                            |
| It can start machining at any location on the workpiece irrespective of presence or absence of edge or through hole | It can start machining from edges only. To initiate W-EDM at an intermediate location on workpiece, a through hole is required. |
| Plunge EDM is commonly used for 3-D contouring, finishing, drilling, deburring, etc.                                | W-EDM is mostly used for high accuracy profile cutting, sample preparation and separating or parting off                        |



# Thermal Processes

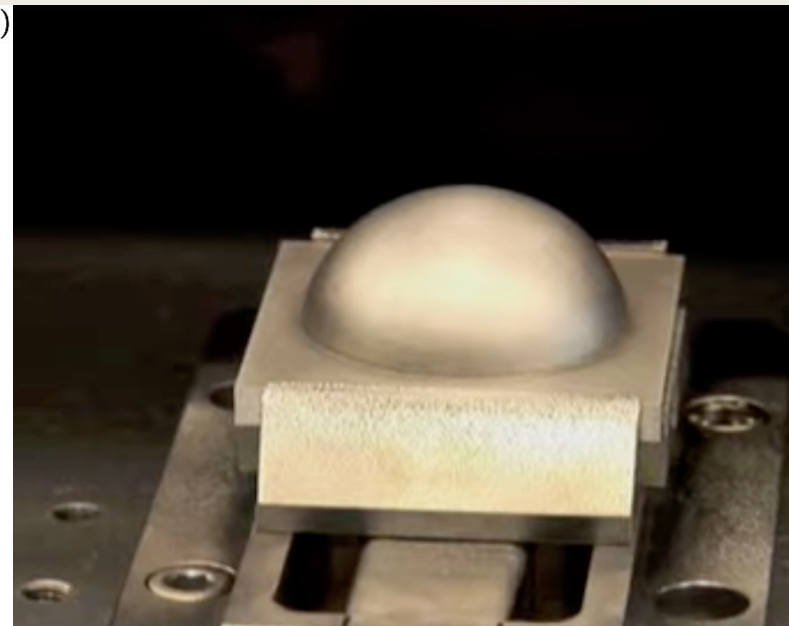
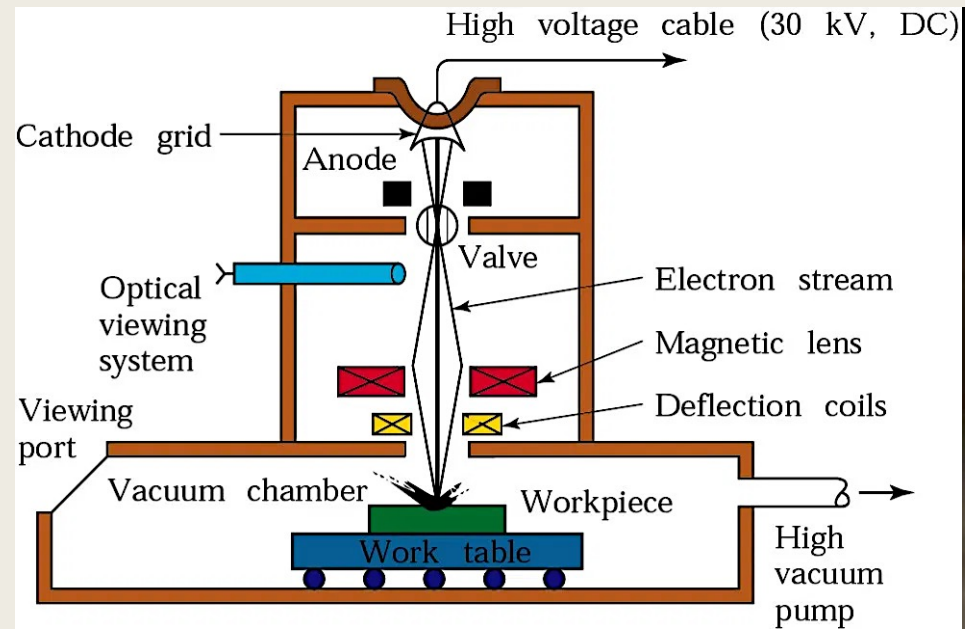




# Thermal Processes



## Electron-Beam Machining





# Thermal Processes



## Electron Beam Machining

vs

## Laser Beam Machining

Both are thermal energy based non-traditional machining processes

Mechanism of material removal (i.e. localized melting and vaporization) is same in both

Both utilize high intensity beam for material removal

No physical contact between tool and workpiece exists in any of these processes

Both are affected by narrow heat affected zone (HAZ) and recast layer.

For both the processes, the kerf width is larger than the beam diameter

Wide variety of pulse modes and durations are also available in both

Computer control system can be integrated with both the processes to facilitate precise control

In both, beam intensity and cross-sectional area can be manipulated to harness desired performance



# Thermal Processes



## Difference between EBM and LBM

| Electron Beam Machining (EBM)                                                                               | Laser Beam Machining (LBM)                                                   |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| A high intensity beam of focused electrons is used                                                          | A high intensity beam of laser (coherent photons) is used                    |
| Always carried out in vacuum (very low pressure) chamber to avoid collision of electrons with air molecules | Does not require vacuum chamber. It can be carried out in open atmosphere    |
| It is suitable for small components                                                                         | Workpiece size does not pose any restriction                                 |
| Time consuming mainly due to creating a low pressure vacuum chamber                                         | Time efficient                                                               |
| X-Ray is generated during EBM process. This possesses serious health concern to the operator.               | No X-Ray is generated in LBM process.                                        |
| Applicable to electrically conductive workpiece materials only                                              | Independent of electrical conductivity of workpiece material.                |
| Optical properties of workpiece surface don't influence EBM process capability                              | LBM process capability relies on the optical properties of workpiece surface |

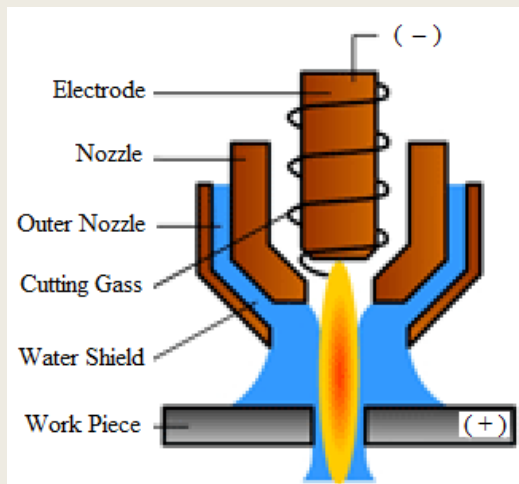


# Thermal Processes

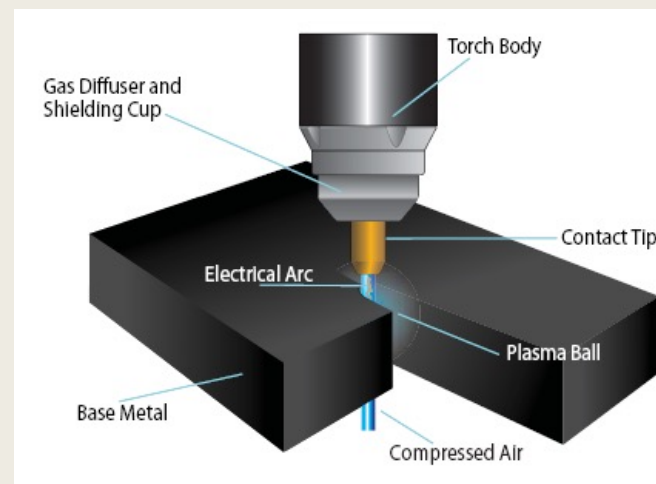


## Plasma Arc Machining

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Plasma Arc cutting mechanism



Plasma Arc cutting process



Machining operation

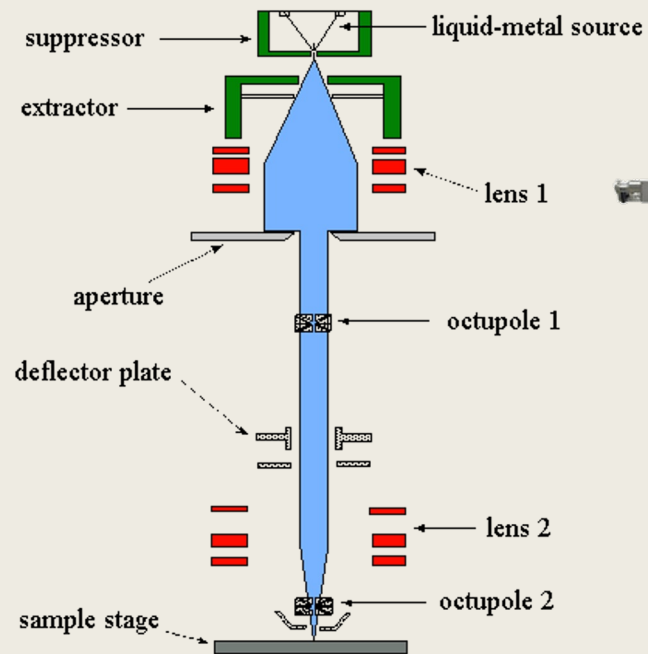




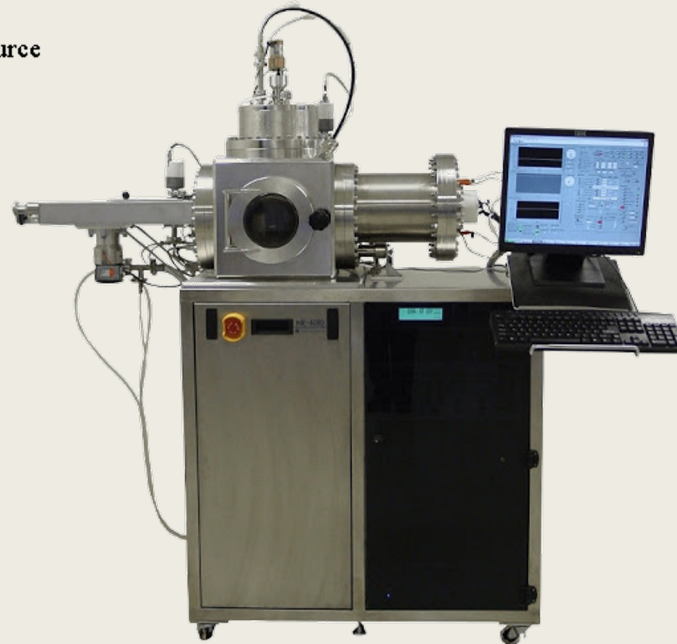
# Thermal Processes



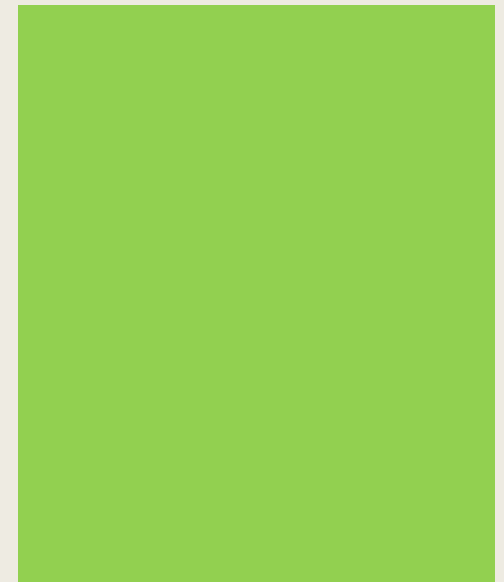
## Ion Beam Machining



Focused ion beam machine structure



Ion beam milling and etching system





# Thermal Processes



## Plasma Arc Machining vs Ion Beam Machining

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Both PAM and IBM are considered as advanced machining processes (AMP) or non-traditional machining (NTM) processes.

Physical contact between a solid tool and workpiece does not exist in either of these two processes. Accordingly, these processes are free from burr formation, progressive wear, mechanical residual stress, etc. However, thermal residual stress may develop in PAM due to thermal cycle.

Process capability for both the cases is independent of mechanical and chemical properties of the workpiece material.

Both the processes can be applied to electrically conductive and non-conductive materials. Note that the arcing fashion for PAM will be different for conductive and non-conductive materials.

Both the plasma beam and ion beam are made of charged particles.

Electrodes are indirectly used in both the processes to generate plasma or ions.

These processes are commonly integrated with computer control system to facilitate precise control.



# Thermal Processes



## Differences between Plasma Arc Machining *and* Ion Beam Machining

| Plasma Arc Machining (PAM)                                                                                            | Ion Beam Machining (IBM)                                                                     |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Thermal energy based advanced machining process.                                                                      | Mechanical or kinetic energy based advanced machining process.                               |
| Mechanism of material removal is a combination of vaporization, and blowing away in molten state                      | Mechanism of material removal is sputtering                                                  |
| A constricted jet of intensely hot plasma is used to supply thermal energy for material removal                       | A beam of high velocity ions is used to transfer momentum for dislodging atoms               |
| Material removal takes place in the form of vapor of the concerned workpiece material.                                | In IBM process, atom (or a cluster of atoms) is removed directly from the workpiece surface. |
| Power density of the plasma jet is relatively higher ( $10^2$ to $10^3$ W/mm <sup>2</sup> ).                          | Power density of ion beam is significantly less ( $10^{-2}$ to $10^0$ W/mm <sup>2</sup> ).   |
| Plasma forming gases (air, argon, nitrogen, hydrogen) must be continuously pumped into the high pressure gas chamber. | A gas (argon) is introduced in a vacuum chamber at a relatively lesser flow rate.            |



# Thermal Processes

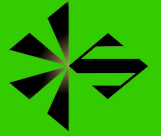


## Differences between Plasma Arc Machining *and* Ion Beam Machining

| Plasma Arc Machining (PAM)                                                                                              | Ion Beam Machining (IBM)                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Does not require a vacuum chamber for its operation. It can be operated in open atmosphere                              | Necessitates a soft vacuum chamber having pressure in the order of $10^{-3}$ to $10^{-5}$ atm                |
| Associated with excessive thermal damage of the workpiece. It produces a wide heat affected zone                        | Thermal damage is mostly neglected (a very narrow zone of atomic level may get affected by impact)           |
| PAM tends to deform the object owing to very high temperature non-uniform heating and subsequent cooling of a wide area | IBM process is free from deformation as heating is mostly negligible                                         |
| Very noisy. Personal protective equipment are indispensably necessary                                                   | IBM process is not noisy                                                                                     |
| Canan be used to cut thick objects (even up to 50 mm)                                                                   | Not suitable for deep cutting. Suitable for micro-fabrication, smoothening and surface contouring            |
| Usually does not generate highly finished surface. Concerned roughness is in the order of $0.5 - 2.0 \mu\text{m}$       | Especially suitable for generating highly finished surfaces having nanometric roughness ( $<10 \text{ nm}$ ) |



## V) Synthesis & Conclusion



# Synthesis and Conclusion

## Non-Traditional Machining Processes

### 1- Chemical & Electrochemical Processes

CHM, PCM, ECM

### 2- Mechanical Processes

AFM, WJM, AWJM, AJM, SB, US

### 3- Thermal Processes

DS-EDM, W-EDM, LBM, EBM, PAM, IBM

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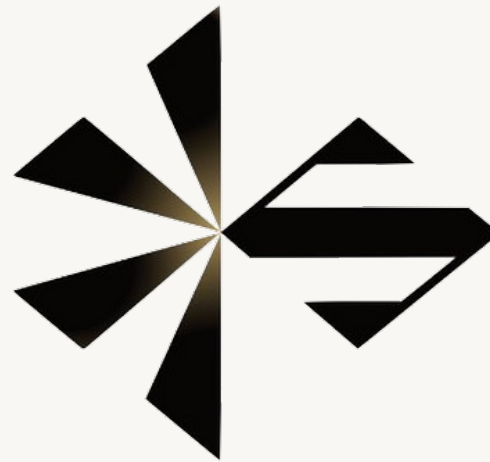
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